1.0 INTRODUCTION

The procedures contained in this manual include all the specifications, instructions, and graphics needed to diagnose body system problems. The diagnostics in this manual are based on the failure condition or symptom being present at time of diagnosis.

Please follow the recommendations below when choosing your diagnostic path.

1. First make sure the DRBIII® is communicating with the appropriate modules: i.e., if the DRBIII® displays a ″No Response″ condition, you must diagnose that first.
2. Read DTC's (diagnostic trouble codes) with the DRBIII®.
3. If no DTC's are present, identify the customer complaint.
4. Once the DTC or customer complaint is identified, locate the matching test in the Table of Contents and begin to diagnose the symptom.

All component location views are in Section 8.0. All connector pinouts are in Section 9.0. All schematics are in Section 10.0.

An asterisk (*) placed before the symptom description indicates a customer complaint.

When repairs are required, refer to the appropriate service manual for the proper removal and repair procedure.

Diagnostic procedures change every year. New diagnostic systems may be added: carryover systems may be enhanced. READ THIS MANUAL BEFORE TRYING TO DIAGNOSE A VEHICLE DIAGNOSTIC TROUBLE CODE. It is recommended that you review the entire manual to become familiar with all the new and changed diagnostic procedures.

This book reflects many suggested changes from readers of past issues. After using this book, if you have any comments or suggestion, please fill out the form in the back of the book, and mail it back to us.

1.1 SYSTEM COVERAGE

This diagnostic procedures manual covers all 2005 Jeep® Wrangler (TJ) vehicles.

1.2 SIX-STEP TROUBLESHOOTING PROCEDURE

Diagnosis of the body system is performed in six basic steps:

1. verification of complaint
2. verification of any related symptoms
3. symptom analysis
4. problem isolation
5. repair of isolated problem
6. verification of proper operation

2.0 IDENTIFICATION OF SYSTEM

The vehicle systems that are part of the ″body″ system are:

- airbag system
- audio
- chime
- electrically heated systems
- instrument cluster
- vehicle communications

3.0 SYSTEM DESCRIPTION AND FUNCTIONAL OPERATION

The body system on the 2005 Jeep® Wrangler (TJ) consists of a combination of modules that communicate over the PCI bus (Programmable Communication Interface multiplex system). Through the PCI bus, information about the operation of vehicle components and circuits is relayed quickly to the appropriate module(s). All modules receive all the information transmitted on the bus even though a module may not require all information to perform its function. It will only respond to messages ″addressed″ to it through a binary coding process. This method of data transmission significantly reduces the complexity of the wiring in the vehicle and the size of wiring harnesses. All of the information about the functioning of all the systems is organized, controlled, and communicated by the PCI bus, which is described in the Vehicle Communication Section of the general information.

3.1 AIRBAG SYSTEM

The Airbag system is designed to provide increased driver and passenger protection if the vehicle is involved in a front-end collision. The airbag system is designed to be used in conjunction with the seat belt system.

Whenever the ignition switch is turned to the Run or Start position, the ACM performs a warning indicator bulb-check via a PCI bus request to the instrument cluster to illuminate the Airbag Warning Indicator. The indicator remains illuminated for 6 to 8 seconds, and then turns off. If the indicator remains off, the ACM has checked the system and found it to be free of discernable malfunctions.

The ACM monitors critical input and output circuits within the airbag system, ensuring the circuits are operating properly. Some circuits are